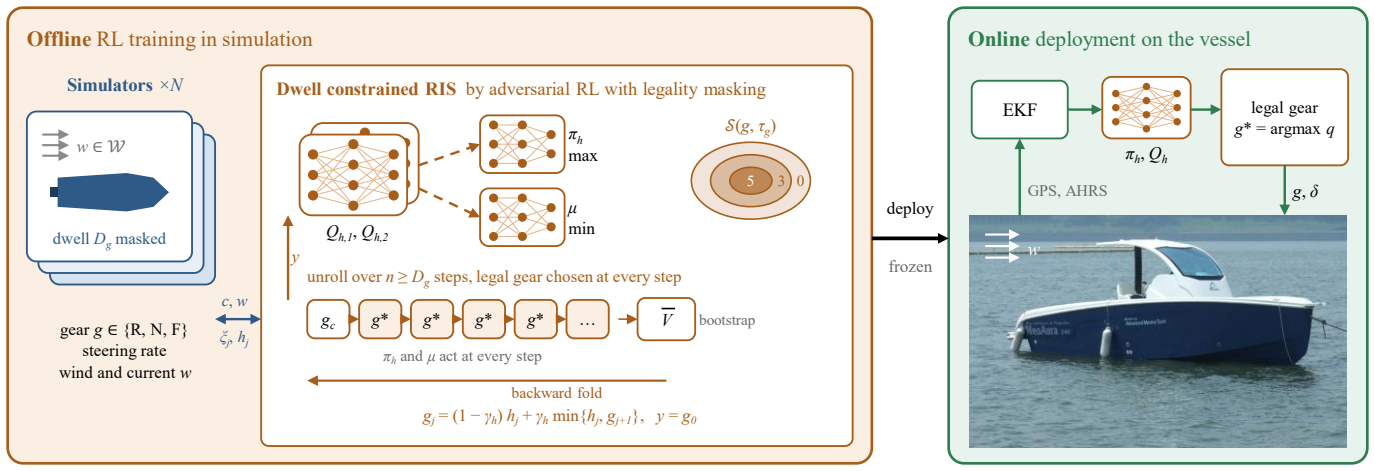

Offline RL training in simulation
Simulators ×N
w ∈ 𝒲
dwell D_g masked
gear g ∈ {R, N, F}
steering rate
wind and current w
c, w
ξ_j, h_j
Dwell constrained RIS by adversarial RL with legality masking
$Q_{h,1}$, $Q_{h,2}$
π_h max
μ min
$\delta(g, \tau_g)$
5
3
0
y
unroll over $n \geq D_g$ steps, legal gear chosen at every step
g_c
g*
g*
g*
g*
…
$\overline{V}$
bootstrap
π_h and μ act at every step
backward fold
$g_j = (1 - \gamma_h)\, h_j + \gamma_h \min\{h_j, g_{j+1}\}$, $y = g_0$
deploy
frozen
Online deployment on the vessel
EKF
π_h, Q_h
legal gear
g* = argmax q
GPS, AHRS
g, δ
w